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The neighbor-joining method: a new method for reconstructing phylogenetic trees

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Abstract

A new method called the neighbor-joining method is proposed for reconstructing phylogenetic trees from evolutionary distance data. The principle of this method is to find pairs of operational taxonomic units (OTUs [= neighbors]) that minimize the total branch length at each stage of clustering of OTUs starting with a starlike tree. The branch lengths as well as the topology of a parsimonious tree can quickly be obtained by using this method. Using computer simulation, we studied the efficiency of this method in obtaining the correct unrooted tree in comparison with that of five other tree-making methods: the unweighted pair group method of analysis, Farris's method, Sattath and Tversky's method, Li's method, and Tateno et al.'s modified Farris method. The new, neighbor-joining method and Sattath and Tversky's method are shown to be generally better than the other methods.

Real-World Impact & Application

Genomic Phylogenetics & Global Virus Tracking: The Neighbor-Joining algorithm is one of the most widely cited algorithms in biological history. Implemented in NCBI BLAST, MEGA, and Nextstrain, it allows molecular biologists to construct evolutionary trees of species from DNA sequences in polynomial time. During the COVID-19 pandemic, variants of this algorithm were used daily worldwide to track emerging viral mutations in real time.